Chapter 9: Differential Equations

School Worksheet: step-wise solutions

20 questions **51** marks **3** levels

Each row is one line of working. The red number in the right margin is what an examiner gives for that line, as in the CBSE marking scheme (½ is half a mark). Write every line that carries a mark. ✓ ½

Level 1

8 marks

1 MCQ — 1 mark

The order and degree of the differential equation $\left[1+\left(\frac{dy}{dx}\right)^2\right]^5 = \frac{d^2y}{dx^2}$ are:

(A) order 1, degree 1 **(B)** order 1, degree 2

(C) **order 2, degree 1** **(D)** order 2, degree 2

Formula / idea: Order = order of the highest derivative; degree = its highest power, once the equation is a polynomial in derivatives.

1 Find the highest order derivative.

$\frac{d^2y}{dx^2} \Rightarrow \text{Order} = 2$

2 The equation is a polynomial in derivatives; power of $\frac{d^2y}{dx^2}$ is 1. ✓ 1

$\text{Degree} = 1$

3 Hence option (C).

Answer **(C) order 2, degree 1**

Watch out: Taking degree 5 or 10 from the power on $\frac{dy}{dx}$; degree is read only from the highest order derivative.

2 MCQ 1 mark

The degree of the differential equation $\frac{d^2y}{dx^2} + \sin\left(\frac{dy}{dx}\right) = 0$ is:

(A) 1 **(B)** 2

(C) 0 **(D)** **not defined**

Formula / idea: Degree is defined only when the equation is a polynomial in its derivatives.

1 Highest order derivative is $\frac{d^2y}{dx^2}$, so order $= 2$.

2 The term $\sin\left(\frac{dy}{dx}\right)$ is not a polynomial in $\frac{dy}{dx}$. ✓ 1

3 So the degree is not defined. Hence option (D).

Answer **(D) not defined**

Watch out: Answering degree 1 by looking only at $\frac{d^2y}{dx^2}$ and ignoring the $\sin$ term.

3 MCQ 1 mark

The integrating factor of the differential equation $x\frac{dy}{dx} - y = x^2$ is:

(A) x

(B) $-x$

(C) $\frac{1}{x}$

(D) $\log x$

Formula / idea: For $\frac{dy}{dx} + Py = Q$, $\text{I.F.} = e^{\int P\,dx}$.

1 Divide by x to get the standard form.

$$\frac{dy}{dx} - \frac{1}{x}y = x$$

2 Compare with $\frac{dy}{dx} + Py = Q$.

$$P = -\frac{1}{x}, \quad Q = x$$

3 Find the integrating factor. ✓ 1

$$\text{I.F.} = e^{\int -\frac{1}{x}dx} = e^{-\log x} = \frac{1}{x}$$

$$e^{-\log x} = e^{\log x^{-1}} = x^{-1}$$

4	Hence option (C).	

Answer **(C)** $\frac{1}{x}$

Watch out: Taking $P = -1$ without first dividing the whole equation by x.

4 MCQ 1 mark

The number of arbitrary constants in the general solution of a differential equation of order 4 is:

(A) 0 **(B)** 2

(C) 3 **(D)** 4

Formula / idea: General solution of an nth order equation has exactly n arbitrary constants.

1	Number of arbitrary constants = order of the equation.	
2	Order is 4, so there are 4 arbitrary constants.	✓ 1
3	Hence option (D).	

Answer **(D)** 4

5 MCQ 1 mark

The general solution of $\frac{dy}{dx} = e^{x+y}$ is:

(A) $e^x + e^{-y} = C$ **(B)** $e^x + e^y = C$

(C) $e^{-x} + e^y = C$ **(D)** $e^{-x} + e^{-y} = C$

Formula / idea: Variable separable: bring all y terms to dy and all x terms to dx, then integrate.

1	Split the exponent. $\frac{dy}{dx} = e^x \cdot e^y$	
2	Separate the variables. $e^{-y}\,dy = e^x\,dx$	
3	Integrate both sides. $-e^{-y} = e^x + C_1$	
4	Rearrange, writing $-C_1 = C$. $e^x + e^{-y} = C$	✓ 1

5 Hence option (A).

Answer **(A)** $e^{x} + e^{-y} = C$

Watch out: Writing $\int e^{-y}dy = e^{-y}$ and losing the minus sign.

6 **Assertion-Reason** 1 mark

Assertion (A): $y = e^{2x}$ is a solution of the differential equation $\frac{d^2y}{dx^2} - 4y = 0$. **Reason (R):** A function is a solution of a differential equation if it satisfies the equation identically.

(A) **Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).**

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

1 Differentiate $y = e^{2x}$ twice.
$\frac{dy}{dx} = 2e^{2x}, \qquad \frac{d^2y}{dx^2} = 4e^{2x}$

2 Substitute in the LHS: it is 0 for every x. So A is true.
$\frac{d^2y}{dx^2} - 4y = 4e^{2x} - 4e^{2x} = 0$

3 R is the definition of a solution. So R is true.

4 A holds exactly because e^{2x} satisfies the equation identically. So R explains A. ✓ 1

5 Hence option (A).

Answer **(A) Both A and R are true and R is the correct explanation of A.**

7 **Short answer** 2 marks

Find the general solution of $\frac{dy}{dx} = \frac{1+y^2}{1+x^2}$.

Formula / idea: $\int \frac{dx}{1+x^2} = \tan^{-1} x + C$

1 Separate the variables. ✓ ½
$\frac{dy}{1+y^2} = \frac{dx}{1+x^2}$

2 Integrate both sides.
$$\int \frac{dy}{1+y^2} = \int \frac{dx}{1+x^2}$$

3 Use the standard integral on each side. ✓ 1
$\tan^{-1} y = \tan^{-1} x + C$

4 Hence the general solution is $\tan^{-1} y = \tan^{-1} x + C$. ✓ ½

Answer $\tan^{-1} y = \tan^{-1} x + C$

Watch out: Forgetting the constant C; a general solution must contain it.

Level 2

18 marks

8 **Short answer** 2 marks

A bacteria sample grows exponentially. The differential equation representing the growth is $\frac{dP}{dt} = kP$, where P is the population at time t. Obtain the general solution and express P as an exponential function of t.

Formula / idea: Separate variables, integrate, then remove the log by taking exponentials.

1 Separate the variables. ✓ ½
$$\frac{dP}{P} = k\,dt$$

2 Integrate both sides. ✓ 1
$$\int \frac{dP}{P} = k\int dt \;\Rightarrow\; \log P = kt + C$$
$P > 0$ for a population, so $|P| = P$.

3 Take exponentials on both sides.
$$P = e^{kt+C} = e^C\, e^{kt}$$

4 Put $t = 0$: $P = e^C$, the initial population. Call it P_0.
$$e^C = P_0$$

5 Hence the population is an exponential function of t. ✓ ½
$$P = P_0\, e^{kt}$$

Answer $P = P_0 e^{kt}$**, where** P_0 **is the population at** $t = 0$

Watch out: Writing $P = e^{kt} + C$; the constant becomes a multiplier e^C, not an added term.

9 **Short answer** 3 marks

Find the particular solution of $\dfrac{dy}{dx} = y\tan x$, given that $y = 1$ when $x = 0$.

Formula / idea: $\displaystyle\int \tan x\,dx = \log|\sec x| + C$

1	Separate the variables. $\dfrac{dy}{y} = \tan x\,dx$	✓ ½
2	Integrate both sides. $\log\lvert y\rvert = \log\lvert\sec x\rvert + C$	✓ 1
3	Put $x = 0, y = 1$. $\log 1 = \log(\sec 0) + C \;\Rightarrow\; 0 = 0 + C$	
4	So the constant is zero. $C = 0$	✓ 1
5	Substitute $C = 0$ and remove logs. Hence the particular solution is $y = \sec x$. $\log\lvert y\rvert = \log\lvert\sec x\rvert \;\Rightarrow\; y = \sec x$ Near $x = 0$, $y > 0$ and $\sec x > 0$.	✓ ½

Answer $y = \sec x$

Watch out: Writing $\displaystyle\int \tan x\,dx = \sec^2 x$; that is the derivative of $\tan x$, not its integral.

10 **Short answer** 3 marks

Solve: $x\dfrac{dy}{dx} + 2y = x^2$ $(x \neq 0)$.

Formula / idea: Linear equation $\dfrac{dy}{dx} + Py = Q$: $y \cdot \text{I.F.} = \displaystyle\int Q \cdot \text{I.F.}\,dx + C$, with $\text{I.F.} = e^{\int P\,dx}$.

1	Divide by x to get the standard linear form. $\dfrac{dy}{dx} + \dfrac{2}{x}y = x, \qquad P = \dfrac{2}{x},\ Q = x$	✓ ½
2	Find the integrating factor. $\text{I.F.} = e^{\int \frac{2}{x}dx} = e^{2\log\lvert x\rvert} = x^2$ $e^{\log\lvert x\rvert^2} = \lvert x\rvert^2 = x^2$	✓ 1
3	Write the solution form $y \cdot \text{I.F.} = \int Q \cdot \text{I.F.}\,dx + C$. $y\,x^2 = \displaystyle\int x \cdot x^2\,dx + C$	✓ ½

4 Integrate.

$$y\,x^2 = \frac{x^4}{4} + C$$

✓ ½

5 Divide by x^2. Hence the general solution is:

$$y = \frac{x^2}{4} + \frac{C}{x^2}$$

✓ ½

Answer $y = \frac{x^2}{4} + \frac{C}{x^2}$

Watch out: Using $P = 2$ instead of $\frac{2}{x}$ because the equation was not divided by x first.

11 **Short answer** 3 marks

Solve the homogeneous differential equation $x\frac{dy}{dx} = x + y$.

Formula / idea: Homogeneous equation $\frac{dy}{dx} = F\left(\frac{y}{x}\right)$: put $y = vx$, so $\frac{dy}{dx} = v + x\frac{dv}{dx}$.

1 Write the equation as $\frac{dy}{dx} = F\left(\frac{y}{x}\right)$; so it is homogeneous.

$$\frac{dy}{dx} = \frac{x+y}{x} = 1 + \frac{y}{x}$$

✓ ½

2 Put $y = vx$.

$$\frac{dy}{dx} = v + x\frac{dv}{dx}$$

✓ ½

3 Substitute and simplify.

$$v + x\frac{dv}{dx} = 1 + v \;\Rightarrow\; x\frac{dv}{dx} = 1$$

✓ ½

4 Separate the variables.

$$dv = \frac{dx}{x}$$

5 Integrate both sides.

$$v = \log|x| + C$$

✓ 1

6 Put back $v = \frac{y}{x}$ and multiply by x. Hence the general solution is:

$$\frac{y}{x} = \log|x| + C \;\Rightarrow\; y = x\log|x| + Cx$$

✓ ½

Answer $y = x\log|x| + Cx$

Watch out: Forgetting to replace v by $\frac{y}{x}$ at the end; the answer must be in x and y.

12 **Short answer** 2 marks

Verify that $y = Ax + \dfrac{B}{x}$ is a solution of $x^2\dfrac{d^2y}{dx^2} + x\dfrac{dy}{dx} - y = 0$.

Formula / idea: A function is a solution if it satisfies the equation identically.

1 Differentiate $y = Ax + Bx^{-1}$ w.r.t. x. ✓ ½

$$\frac{dy}{dx} = A - \frac{B}{x^2}$$

2 Differentiate again. ✓ ½

$$\frac{d^2y}{dx^2} = \frac{2B}{x^3}$$

3 Substitute in the LHS. ✓ ½

$$\text{LHS} = x^2 \cdot \frac{2B}{x^3} + x\left(A - \frac{B}{x^2}\right) - \left(Ax + \frac{B}{x}\right)$$

4 Simplify.

$$\text{LHS} = \frac{2B}{x} + Ax - \frac{B}{x} - Ax - \frac{B}{x} = 0 = \text{RHS}$$

5 Hence $y = Ax + \dfrac{B}{x}$ is a solution of the given equation. ✓ ½

Answer **LHS $= 0 =$ RHS, so $y = Ax + \dfrac{B}{x}$ is a solution.**

Watch out: Sign slip: the derivative of $\dfrac{B}{x}$ is $-\dfrac{B}{x^2}$, and of that, $+\dfrac{2B}{x^3}$.

13 **Short answer** 2 marks

Find the general solution of $\dfrac{dy}{dx} = \sin^{-1} x$.

Formula / idea: Integration by parts: $\displaystyle\int u\,v\,dx = u\int v\,dx - \int\left(u'\int v\,dx\right)dx$.

1 Separate the variables and integrate both sides. ✓ ½

$$dy = \sin^{-1} x\,dx \;\Rightarrow\; y = \int \sin^{-1} x \cdot 1\,dx$$

2 Integrate by parts: $\sin^{-1} x$ first function, 1 second function. ✓ ½

$$y = x\sin^{-1} x - \int \frac{x}{\sqrt{1-x^2}}\,dx$$

3 Put $1 - x^2 = t$, so $x\,dx = -\dfrac{dt}{2}$. ✓ ½

$$\int \frac{x}{\sqrt{1-x^2}}\,dx = -\frac{1}{2}\int t^{-1/2}\,dt = -\sqrt{1-x^2}$$

4 Substitute back. Hence the general solution is: ✓ ½

$y = x\sin^{-1}x + \sqrt{1-x^2} + C$

Answer $y = x\sin^{-1}x + \sqrt{1-x^2} + C$

Watch out: Sign error: $-\int \frac{x}{\sqrt{1-x^2}}dx$ gives $+\sqrt{1-x^2}$, not $-\sqrt{1-x^2}$.

14 **Short answer** 3 marks

Find the particular solution of the differential equation $\frac{dy}{dx} - 2xy = 3x^2e^{x^2}$, given $y(0) = 5$.

Formula / idea: Linear equation: $y \cdot \text{I.F.} = \int Q \cdot \text{I.F.}\, dx + C,\ \text{I.F.} = e^{\int P\,dx}$.

1 Compare with $\frac{dy}{dx} + Py = Q$. ✓ ½

$P = -2x, \qquad Q = 3x^2e^{x^2}$

2 Find the integrating factor. ✓ ½

$\text{I.F.} = e^{\int -2x\,dx} = e^{-x^2}$

3 Write the solution form. ✓ ½

$y\,e^{-x^2} = \int 3x^2e^{x^2}\cdot e^{-x^2}\,dx + C = \int 3x^2\,dx + C$

4 Integrate. ✓ ½

$y\,e^{-x^2} = x^3 + C$

5 Put $x = 0, y = 5$. ✓ ½

$5\cdot e^0 = 0 + C \ \Rightarrow\ C = 5$

6 Substitute $C = 5$ and multiply by e^{x^2}. Hence the particular solution is: ✓ ½

$y = (x^3 + 5)\,e^{x^2}$

Answer $y = (x^3 + 5)\,e^{x^2}$

Watch out: Taking $\text{I.F.} = e^{x^2}$; here $P = -2x$, so the I.F. is e^{-x^2}.

Level 3

25 marks

15 **Long answer** 5 marks

Solve: $x\,dy - y\,dx = \sqrt{x^2+y^2}\,dx$.

Formula / idea: Homogeneous equation: put $y = vx$, $\dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$; and $\displaystyle\int \frac{dv}{\sqrt{1+v^2}} = \log\left|v+\sqrt{1+v^2}\right|$.

1 Rewrite in the form $\dfrac{dy}{dx} = F(x,y)$. ✓ ½

$$\frac{dy}{dx} = \frac{y+\sqrt{x^2+y^2}}{x}$$

2 Check: $F(\lambda x, \lambda y) = \lambda^0 F(x,y)$, so the equation is homogeneous. ✓ ½

$$F(\lambda x,\lambda y) = \frac{\lambda y + \lambda\sqrt{x^2+y^2}}{\lambda x} = F(x,y)$$

3 Put $y = vx$. ✓ ½

$$\frac{dy}{dx} = v + x\frac{dv}{dx}$$

4 Substitute, taking $x > 0$. ✓ ½

$$v + x\frac{dv}{dx} = \frac{vx+\sqrt{x^2+v^2x^2}}{x} = v + \sqrt{1+v^2}$$

5 Cancel v on both sides. ✓ ½

$$x\frac{dv}{dx} = \sqrt{1+v^2}$$

6 Separate the variables. ✓ ½

$$\frac{dv}{\sqrt{1+v^2}} = \frac{dx}{x}$$

7 Integrate both sides; write the constant as $\log C$. ✓ 1

$$\log\left|v+\sqrt{1+v^2}\right| = \log|x| + \log C$$

8 Combine the logs. ✓ ½

$$v + \sqrt{1+v^2} = Cx$$

9 Put $v = \dfrac{y}{x}$ and multiply by x. Hence the general solution is: ✓ ½

$$\frac{y+\sqrt{x^2+y^2}}{x} = Cx \;\Rightarrow\; y + \sqrt{x^2+y^2} = Cx^2$$

Answer $y + \sqrt{x^2+y^2} = Cx^2$

Watch out: Writing $\sqrt{x^2+v^2x^2} = x + vx$; the correct step is $x\sqrt{1+v^2}$.

16 **Short answer** 3 marks

Find the particular solution of $(1+x^2)\dfrac{dy}{dx} + 2xy = \dfrac{1}{1+x^2}$, given $y = 0$ when $x = 1$.

Formula / idea: Linear equation: $y \cdot \text{I.F.} = \displaystyle\int Q \cdot \text{I.F.}\, dx + C,\ \text{I.F.} = e^{\int P\, dx}$.

1 Divide by $(1+x^2)$ to get the standard form. ✓ ½

$$\frac{dy}{dx} + \frac{2x}{1+x^2}\,y = \frac{1}{(1+x^2)^2}$$

2 Find the integrating factor. ✓ ½

$$\text{I.F.} = e^{\int \frac{2x}{1+x^2}\,dx} = e^{\log(1+x^2)} = 1+x^2$$

3 Write the solution form. ✓ ½

$$y(1+x^2) = \int \frac{1}{(1+x^2)^2} \cdot (1+x^2)\,dx + C = \int \frac{dx}{1+x^2} + C$$

4 Integrate. ✓ ½

$$y(1+x^2) = \tan^{-1} x + C$$

5 Put $x = 1, y = 0$. ✓ ½

$$0 = \tan^{-1} 1 + C = \frac{\pi}{4} + C \ \Rightarrow\ C = -\frac{\pi}{4}$$

6 Hence the particular solution is: ✓ ½

$$y(1+x^2) = \tan^{-1} x - \frac{\pi}{4}$$

Answer $y(1+x^2) = \tan^{-1} x - \dfrac{\pi}{4}$

Watch out: Not dividing the RHS by $(1+x^2)$ too, so Q is taken as $\dfrac{1}{1+x^2}$ instead of $\dfrac{1}{(1+x^2)^2}$.

17 **Long answer** 5 marks

Find the particular solution of $x \sin\left(\dfrac{y}{x}\right)\dfrac{dy}{dx} + x - y\sin\left(\dfrac{y}{x}\right) = 0$, given $y = \dfrac{\pi}{2}$ when $x = 1$.

Formula / idea: Homogeneous equation: put $y = vx$, $\dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$, separate, integrate, put back $v = \dfrac{y}{x}$.

1 Make $\dfrac{dy}{dx}$ the subject. ✓ ½

$$\frac{dy}{dx} = \frac{y\sin\left(\frac{y}{x}\right) - x}{x\sin\left(\frac{y}{x}\right)}$$

2 RHS is a function of $\dfrac{y}{x}$ only, so the equation is homogeneous. ✓ ½

$$\frac{dy}{dx} = \frac{y}{x} - \frac{1}{\sin\left(\frac{y}{x}\right)}$$

3 Put $y = vx$.
$\frac{dy}{dx} = v + x\frac{dv}{dx}$
✓ ½

4 Substitute and cancel v.
$v + x\frac{dv}{dx} = v - \frac{1}{\sin v} \Rightarrow x\frac{dv}{dx} = -\frac{1}{\sin v}$
✓ ½

5 Separate the variables.
$\sin v\, dv = -\frac{dx}{x}$
✓ ½

6 Integrate both sides; multiply by -1 and rename the constant C.
$-\cos v = -\log|x| + C_1 \Rightarrow \cos v = \log|x| + C$
✓ 1

7 Put back $v = \frac{y}{x}$.
$\cos\left(\frac{y}{x}\right) = \log|x| + C$
✓ ½

8 Put $x = 1, y = \frac{\pi}{2}$.
$\cos\frac{\pi}{2} = \log 1 + C \Rightarrow 0 = 0 + C \Rightarrow C = 0$
✓ ½

9 Hence the particular solution is:
$\cos\left(\frac{y}{x}\right) = \log|x|$
✓ ½

Answer $\cos\left(\frac{y}{x}\right) = \log|x|$

Watch out: Writing $\int \sin v\, dv = \cos v$; it is $-\cos v$.

18 Short answer — 3 marks

Solve the differential equation: $x^2\, dy + y(x+y)\, dx = 0$.

Formula / idea: Homogeneous equation: put $y = vx$; use partial fractions $\frac{1}{v(v+2)} = \frac{1}{2}\left(\frac{1}{v} - \frac{1}{v+2}\right)$.

1 Write $\frac{dy}{dx}$; numerator and denominator are both of degree 2, so it is homogeneous.
$\frac{dy}{dx} = -\frac{xy + y^2}{x^2}$
✓ ½

2 Put $y = vx$, $\frac{dy}{dx} = v + x\frac{dv}{dx}$.
$v + x\frac{dv}{dx} = -(v + v^2)$
✓ ½

3 Simplify and separate the variables.
$x\frac{dv}{dx} = -(v^2 + 2v) \Rightarrow \frac{dv}{v(v+2)} = -\frac{dx}{x}$
✓ ½

4 Use partial fractions and integrate.

$$\frac{1}{2}\int\left(\frac{1}{v}-\frac{1}{v+2}\right)dv=-\int\frac{dx}{x}$$

5 Integrate; write the constant as $\log|C_1|$. ✓ 1

$$\frac{1}{2}\log\left|\frac{v}{v+2}\right|=-\log|x|+\log|C_1|$$

6 Multiply by 2 and remove logs, writing $C=\pm C_1^2$.

$$\frac{v}{v+2}=\frac{C}{x^2}$$

7 Put $v=\dfrac{y}{x}$. Hence the general solution is: ✓ ½

$$\frac{y}{y+2x}=\frac{C}{x^2}\;\Rightarrow\;x^2y=C(y+2x)$$

Answer $x^2y=C(y+2x)$

Watch out: Dropping the factor $\frac{1}{2}$ from the partial fractions, which changes x^2 to x in the answer.

19 Long answer — 5 marks

Solve: $(\tan^{-1}y-x)\,dy=(1+y^2)\,dx$.

Formula / idea: Linear in x: $\dfrac{dx}{dy}+P_1x=Q_1$ gives $x\cdot\text{I.F.}=\displaystyle\int Q_1\cdot\text{I.F.}\,dy+C,\ \text{I.F.}=e^{\int P_1dy}$.

1 Write $\dfrac{dx}{dy}$, since the equation is linear in x. ✓ ½

$$\frac{dx}{dy}=\frac{\tan^{-1}y-x}{1+y^2}$$

2 Rearrange to the form $\dfrac{dx}{dy}+P_1x=Q_1$. ✓ ½

$$\frac{dx}{dy}+\frac{1}{1+y^2}\,x=\frac{\tan^{-1}y}{1+y^2}$$

3 Find the integrating factor. ✓ 1

$$\text{I.F.}=e^{\int\frac{dy}{1+y^2}}=e^{\tan^{-1}y}$$

4 Write the solution form. ✓ ½

$$x\,e^{\tan^{-1}y}=\int\frac{\tan^{-1}y}{1+y^2}\,e^{\tan^{-1}y}\,dy+C$$

5 Call this integral I. Put $\tan^{-1}y=t$, so $\dfrac{dy}{1+y^2}=dt$. ✓ ½

$$I=\int\frac{\tan^{-1}y}{1+y^2}\,e^{\tan^{-1}y}\,dy=\int t\,e^t\,dt$$

6 Integrate by parts. ✓ 1

$$I = t\,e^{t} - \int e^{t}\,dt = (t-1)\,e^{t} = (\tan^{-1} y - 1)\,e^{\tan^{-1} y}$$

7 Substitute I back. ✓ ½

$$x\,e^{\tan^{-1} y} = (\tan^{-1} y - 1)\,e^{\tan^{-1} y} + C$$

8 Divide by $e^{\tan^{-1} y}$. Hence the general solution is: ✓ ½

$$x = \tan^{-1} y - 1 + C\,e^{-\tan^{-1} y}$$

Answer $x = \tan^{-1} y - 1 + Ce^{-\tan^{-1} y}$

Watch out: Trying to solve it as $\frac{dy}{dx}$; it is linear only in x, so use $\frac{dx}{dy}$.

20 Case study — 4 marks

Case Study: In a laboratory culture, the number of bacteria P increases at a rate proportional to the number present. The count of 1,00,000 bacteria increases by 10% in 2 hours.

(i) Write the differential equation modelling this situation and state its order and degree. — 1 mark

(ii) Solve it to express P in terms of t and find the constant k. — 1 mark

(iii) In how many hours will the count reach 2,00,000? — 2 marks

Formula / idea: Rate proportional to amount: $\frac{dP}{dt} = kP$ gives $P = P_0 e^{kt}$.

(i) — 1 mark

1 Rate of increase is proportional to P, with constant $k > 0$. ✓ ½

$$\frac{dP}{dt} = kP$$

2 Highest derivative is $\frac{dP}{dt}$, with power 1. ✓ ½

$$\text{Order} = 1, \quad \text{Degree} = 1$$

Answer (i) $\frac{dP}{dt} = kP$**; order 1, degree 1**

(ii) — 1 mark

1 Separate and integrate.

$$\frac{dP}{P} = k\,dt \;\Rightarrow\; \log P = kt + C$$

2 At $t = 0$, $P = 100000$, so $C = \log 100000$. ✓ ½

$$\log \frac{P}{100000} = kt \;\Rightarrow\; P = 100000\,e^{kt}$$

3 At $t = 2$, $P = 100000 + 10\%$ of $100000 = 110000$.

$$110000 = 100000\,e^{2k} \;\Rightarrow\; e^{2k} = 1.1$$

4 Take log on both sides.

$2k = \log 1.1 \Rightarrow k = \frac{1}{2}\log 1.1$ ✓ ½

Answer (ii) $P = 100000\,e^{kt}$, $k = \frac{1}{2}\log 1.1\ (\approx 0.0477)$

(iii)

2 marks

1 Put $P = 200000$.

$200000 = 100000\,e^{kt} \Rightarrow e^{kt} = 2$ ✓ ½

2 Take log on both sides.

$kt = \log 2$ ✓ ½

3 Put $k = \frac{1}{2}\log 1.1$.

$t = \frac{\log 2}{k} = \frac{2\log 2}{\log 1.1}$ ✓ ½

4 Use $\log 2 \approx 0.6931$, $\log 1.1 \approx 0.0953$. Hence the count reaches 2,00,000 in:

$t \approx \frac{2 \times 0.6931}{0.0953} \approx 14.5$ hours ✓ ½

Answer (iii) $t = \frac{2\log 2}{\log 1.1} \approx 14.5$ **hours**

Watch out: Taking $P = 10000$ (the increase) instead of $P = 110000$ (the new count) at $t = 2$.